

ECONOMICS

Breakeven Analysis

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Break-Even Point Analysis

A decision-making aid that enables a manager to determine whether a particular volume of sales will result in **losses** or **profits**

Basic Concepts

Variable costs are costs that change with changes in production levels or sales.

Examples include: *Costs of materials used in the production of the goods*

Fixed costs remain roughly the same regardless of sales/output levels.

Examples include: *Rent, Insurance and Wages*

Revenue is the total income received.

Profit is the money you have after subtracting fixed and variable cost from revenue.

What it can be used for?

- Monthly expenses- use it to see if your income is more than your expenses
- Determine minimum price product can be sold for
- Determine optimum price product can be sold for
- Calculate effects of marketing programs on price

The Break-Even Point (BEP)

$$P(X) = F + V(X)$$

Revenue *Total Costs*

Where:

- F = fixed costs
- V = variable costs per unit
- X = volume of output (in units)
- P = price per unit

The Break-Even Point (BEP)

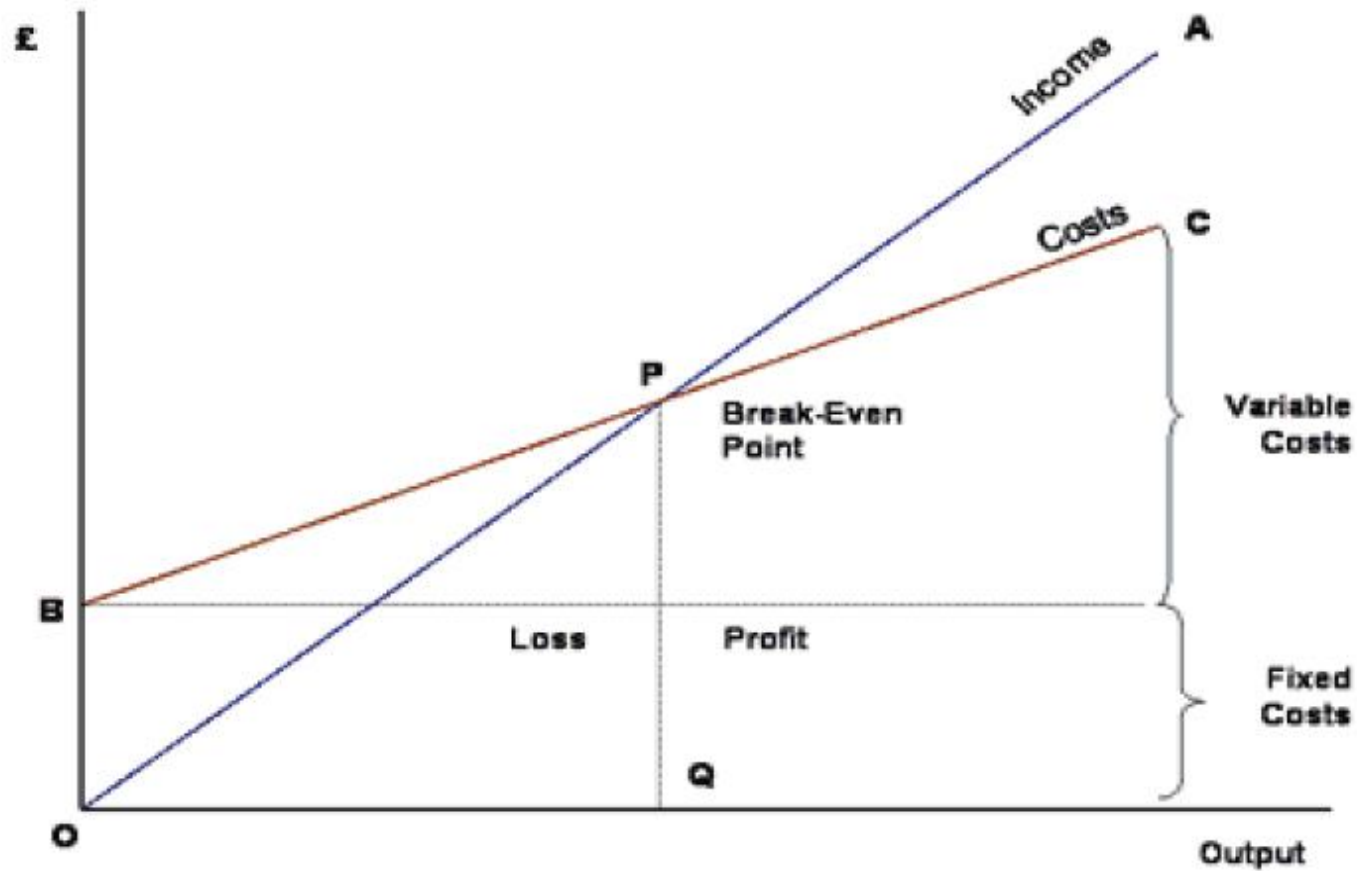
If we rearrange the formula where the breakeven is **X** then the formula look like this

$$\mathbf{X = F / (P - V)}$$

*Break-Event
Point*

This formula says that the breakeven point is where the number of sales needed to make the cost equal to the revenue.

The Break-Even Point Chart



Sample Problem # 1

Let's say you own a business selling burgers

It costs **USD 7.00** to make one burger **That's your V or Variable cost!**

You sell each burger for **USD 12.50**.
That's your P or price per unit!

Your cost for rent, utilities, overhead, etc... is
USD 100,000 per month.
That's your F or fixed cost!

Sample Problem # 1 – Solution

Given

:

$V = \text{USD } 7, P = \text{USD } 12.50$

$F = \text{USD } 100,000$

- Solution:
- $X = F / (P - V)$
- $X = 100,000 / (12.50 - 7) \quad X = 100,000 / (5.5)$
- $X = 18,181.818$
- $X \approx 18,182$

To breakeven you would need to sell **18182** burgers.

Sample Problem # 2

Try out this problem for your self

- You own a lemonade stand.
- It costs you **USD 1.50** to make cup of lemonade
- You sell your lemonade for **USD 5.00**.
- It cost you **USD5,000** to rent for the space of your lemonade stand.
- How many cups of lemonade do you have to sell to breakeven?

Solve now

Sample Problem # 2 – Solution

Given

:

$$V = \text{USD } 1.50 \quad P = \text{USD } 5.00$$

$$F = \text{USD } 5,000$$

Solution:

$$X = F / (P - V)$$

$$X = 5000 / (5 - 1.50)$$

$$X = 5000 / (3.5)$$

$$X = 1,428.57$$

$$X \approx 1,429$$

You would need to sell 1,429 cups of lemonade to breakeven.

Sample Problem # 3 – Solution

Bella owns a small bakery called **Bella's Bakes** that specializes in premium cupcakes. She is considering launching a new line of **gluten-free cupcakes** and wants to assess the financial viability before committing to production.

- **Fixed Costs: \$12,000**
(Includes rent, equipment leasing, marketing for launch)
- **Variable Cost per Unit: \$1.80**
(Ingredients, packaging, and direct labor per cupcake)
- **Selling Price per Unit: \$4.50**

Sample Problem # 3 – Solution

$$\begin{aligned}\text{Break-even Point (units)} &= \frac{\text{Fixed Costs}}{\text{Selling Price} - \text{Variable Cost}} = \frac{12,000}{4.50 - 1.80} \\ &= \frac{12,000}{2.70} = 4,444.44 \approx 4,445 \text{ cupcakes}\end{aligned}$$

Sample Problem # 4 – Solution

EcoBottle Pvt. Ltd. is a startup that manufactures biodegradable water bottles. The company plans to expand operations by opening a new production unit to meet increasing demand and reduce per-unit costs. Management must understand when the new unit will become profitable.

"How many units must we sell to break even if we expand production, and is the expansion financially viable under different pricing and cost scenarios?"

Element	Value
Fixed Costs (monthly)	\$150,000
Variable Cost per Bottle	\$1.80
Planned Selling Price	\$4.50
Production Capacity	100,000 bottles/month
Expected Monthly Sales	75,000 bottles
Tax and Interest	Ignored for break-even analysis

Sample Problem # 4 – Solution

Step 1: Calculate Contribution Margin (CM)

$$\begin{aligned}\text{Contribution Margin per unit} &= \text{Selling Price} - \text{Variable Cost} \\ &= 4.50 - 1.80 = \$2.70\end{aligned}$$

Step 2: Calculate Break-even Point in Units

$$\begin{aligned}\text{Break-even Point (units)} &= \frac{\text{Fixed Costs}}{\text{Contribution Margin per unit}} \\ &= \frac{150,000}{2.70} \approx 55,556 \text{ bottles}\end{aligned}$$

Sample Problem # 4 – Solution

Step 3: Break-even in Revenue

$$\text{Break-even Revenue} = 55,556 \times 4.50 = \$250,002$$

Step 4: Scenario Analysis

Scenario A: Selling Price Drop to \$4.00

If competition forces EcoBottle to reduce the price to \$4.00:

- New CM = $\$4.00 - \$1.80 = \$2.20$
- New BEP = $\$150,000 / 2.20 = 68,182$ bottles

Impact: +12,626 more bottles must be sold to break even

Scenario B: Increase in Variable Costs to \$2.20 (e.g., material inflation)

- CM = $\$4.50 - \$2.20 = \$2.30$
- BEP = $\$150,000 / 2.30 \approx 65,217$ bottles

Impact: +9,661 more bottles required to break even

Sample Problem # 4 – Solution

Scenario C: Dual Impact — Price cut to \$4.00 and cost increase to \$2.20

- $CM = \$4.00 - \$2.20 = \$1.80$
- $BEP = \$150,000 / 1.80 = 83,333 \text{ bottles}$

Impact: +27,777 bottles — becomes financially risky

Step 5: Graphical Analysis

A break-even chart would show:

- X-axis: Units sold (0–100,000)
- Y-axis: Revenue and Cost (\$)
- Lines for:

Step 6: Margin of Safety

$$\begin{aligned} \text{Margin of Safety} &= \text{Expected Sales} - \text{Break-even Sales} \\ &= 75,000 - 55,556 = 19,444 \text{ bottles} \end{aligned}$$

- **Total Revenue:** starting at zero, slope = price
- **Total Cost:** starting at fixed cost, slope = variable cost
- The intersection = break-even point (55,556 units)

Sample Problem # 4 – Solution

Step 7: Strategic Recommendations

Feasibility

- With a BEP of 55,556 bottles and a capacity of 100,000 bottles, break-even is achievable.
- A price cut or cost increase substantially raises the break-even point — need cost control.

Recommendations

1. Negotiate supplier contracts to maintain raw material cost below \$2.00.
2. Monitor pricing strategy — price should not fall below \$4.20 if margins are to remain strong.
3. Introduce multi-unit packages (e.g., 6-pack eco bundles) to stabilize revenue.
4. Use margin of safety as a risk cushion — ensure preorders are at least 70,000 units.
5. Model flexible scenarios monthly for dynamic pricing and volume strategies.

Sample Problem # 1 – Production Possibility Frontier

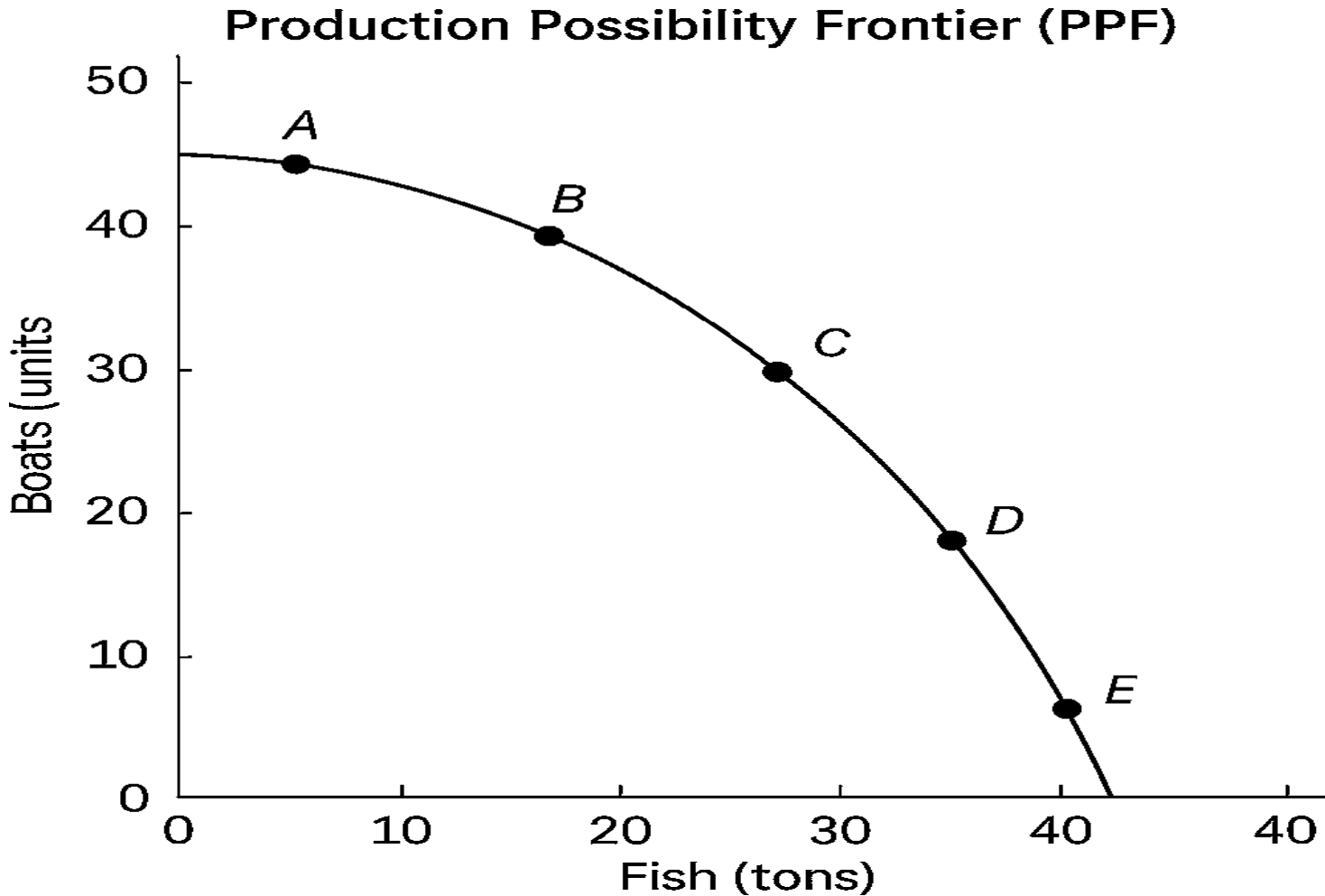
Imagine a small island nation called EconIsland. The island has limited resources land, labor, and tools, which can be used to produce two main goods 1) Fish 🐟 (food for survival) 2) Boats 🚤 (for transportation and trade). The leaders of EconIsland must decide how to use their resources to balance both goods. They can either focus all efforts on catching fish, building boats, or do a mix of both.

Option	Fish (tons)	Boats (units)
A	0	50
B	10	45
C	20	35
D	30	20
E	40	0

Sample Problem # 1 – Production Possibility Frontier

1. Draw the PPF Curve using the data.
2. What do points on, inside, and outside the PPF mean?
3. Calculate the opportunity cost of moving from point B to C.
4. Is the opportunity cost constant or increasing? Why?
5. Suppose a new fishing net technology is introduced. How would that affect the PPF?
6. If the island is producing at point D but the population is starving, what should they do?

Sample Problem # 1 – Production Possibility Frontier



Sample Problem # 1 – Production Possibility Frontier

Opportunity Cost (What You Give Up)

Let's say the island moves from B (10 fish, 45 boats) to C (20 fish, 35 boats).

- It gains 10 tons of fish
- But it loses 10 boats

So, the opportunity cost of 10 fish = 10 boats

➡ This means for every extra fish, they must give up 1 boat.

Now try from C to D:

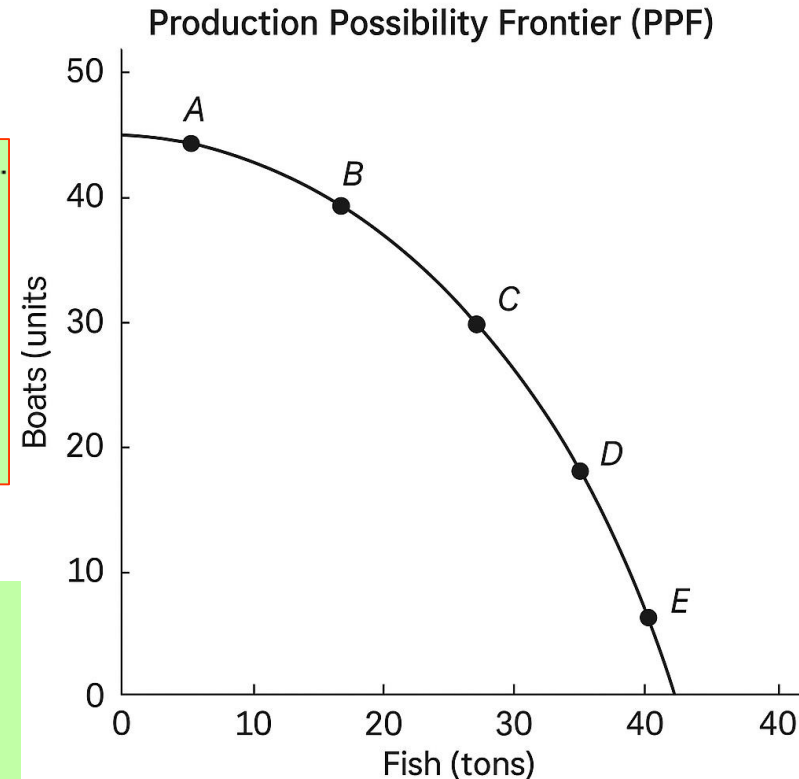
- Gain = 10 fish
- Loss = 15 boats

➡ Now each extra fish costs 1.5 boats

This shows:

📌 The more you produce of one good, the more you have to sacrifice of the other.

This is called increasing opportunity cost, and that's why the curve is bowed out.



Limitations of BEP Analysis

- Assumes that sales prices are constant at all levels of output.
- Assumes production and sales are the same.
- Break even charts may be time consuming to prepare.
- It can only apply to a single product or single mix of products.

Conclusion

- A Breakeven Analysis is a simple tool to use to determine if you have priced your product correctly.
- A Breakeven Analysis helps you calculate how much you need to sell before you begin to make a profit. You can also see how fixed costs, price, volume, and other factors affect your net profit.